Exam Causal Machine Learning

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You are given 3 hours to work on this exam. It is compulsory to hand in the question sheet at the end of the exam.

1. Suppose that we are interested in the average causal effect of a (dichotomous) treatment on an outcome. Suppose that we have data from an observational study, which contains a large collection of covariates or features that includes all confounders needed to adjust for confounding, but possibly more. Someone proposes to train random forest regression to predict outcome in function of covariates, separately in treated and untreated individuals. Based on the results, s/he then averages the difference between the predictions on treatment and without treatment across all individuals in the sample and proposes this as an estimate of the average causal effect:

   (a) (2 points) Do you expect this method (i.e., random forests regression) to make an appropriate selection of confounders? Why (not)?

   (b) (2 points) Do you expect that a valid standard error can be obtained by replicating the above procedure using the bootstrap? Why (not)? A brief answer can suffice; no detailed explanation is expected.

   (c) (2 points) Do you see better ways of estimating the average causal effect along with a standard error? How? It suffices to refer to one specific procedure without explaining the details of how it works. However, I do expect that you give some insight why it is better than the above sketched procedure.

2. Consider assumption-lean logistic regression of a dichotomous outcome Y on a dichotomous treatment A, adjusting for baseline characteristics L. Suppose that the treatment is randomly assigned so that everyone has a 50% chance to be treated.

   (a) (2 points) Then explain what is the estimand targeted by assumption-lean logistic regression (adjusting for L) in the context of such randomized experiment. Try to be as clear as possible.

   (b) (2 points) Do you expect this estimand to have the same magnitude as the estimand targeted by assumption-lean logistic regression when not adjusting for L? Explain.

   (c) (1 points) Would it be justified to substitute $E(A|L)$ in the assumption-lean logistic regression procedure by (a) 0.5; (b) the sample average $\sum_{i=1}^{n} A_i/n$; (c) a random forest prediction (based on features L)? Mark those that apply.

3. (a) (3 points) What do you view as key advantages of the use of orthogonal learners (like DR-learner, R-learner) for the estimation of conditional average treatment effects over non-orthogonal learners (like T-learner)?

   (b) (1 points) Are these advantages limited to observational studies, or are some also relevant for randomized experiments?

   (c) (2 points) Suppose you were given a learner $\psi(L)$ for $E(Y^1|L)$ (in other words, suppose that $\psi(L)$ is known). Would it be valid to estimate the corresponding mean squared error

$$E\left[\left\{Y^1 - \psi(L)\right\}^2\right]$$

as the sample average of $\{Y - \psi(L)\}^2$ among treated individuals? Why (not)?

(d) (2 points) Do you see a way of estimating this mean squared error using debiased machine learning, when data are available on a binary treatment A, an outcome Y and a collection of covariates L that suffices to adjust for confounding (you may also assume the consistency assumption holds that $Y = Y^1$ when $A = 1$)? How?

4. Researchers want to investigate the causal effect of diabetes on Alzheimer's disease. The association between diabetes and Alzheimer's disease is confounded by possibly unmeasured factors U. Suppose now that data are available on the HLA-DR3 genotype G (which you may view as a continuous score), diabetes X and Alzheimer's disease Y. Carriers of one specific variant of the HLA-DR3 genotype are more likely to develop diabetes. Suppose furthermore that the HLA-DR3 genotype can only affect Alzheimer's disease by influencing the risk of diabetes. Finally, assume that by Mendelian inheritance, the HLA-DR3 genotype can be viewed as being 'randomised'.

   (a) (2 points) Draw a causal diagram according to this situation.

   (b) (2 points) Do you see ways of testing whether diabetes affects the risk of Alzheimer's disease, despite the presence of unmeasured confounding? How?

   (c) Let Y^x denote the counterfactual cardiovascular disease status that would be seen for given person if his/her diabetes status were x.

      i. (2 points) Use SWIGs, based on your causal diagram, to show that $Y^x \perp\!\!\!\perp G$.

      ii. (2 points) What is the practical meaning of this independence assumption? In other words, what is the key information it is telling us?

      iii. (2 points) Is this assumption testable? In other words, can it be verified based on the data? If yes, how? If not, why?

   (d) Suppose now that we data on the individual's family history of Alzheimer's disease H (where H may be a vector encoding multiple aspects of family history), and assume that this is associated with both genotype (indirectly, via the ancestral genotypes) as well as the individual's risk of Alzheimer's disease. Explain how you could use causal machine learning

      i. (3 points) to estimate the effect of diabetes on the risk of Alzheimer's disease.

      ii. (2 points) to test whether diabetes affects the risk of Alzheimer's disease.

   (e) (4 points) Suppose now that data are only collected among individuals of at least 80 years old (because the risk of Alzheimer's disease is larger in that population). Reason, using causal diagrams, whether the fact that people may die before they are 80 years old (and that we only collect data for people who are 80 years or older) may induce selection bias. For this, extend the causal diagram with a node to indicate the live status (alive or death) of the person at the start of the study, Next, reason using the causal diagram whether there may be selection bias in the following cases:

      i. mortality is only affected by diabetes X;

      ii. mortality is only affected by Alzheimer's disease Y;

      iii. mortality is affected by diabetes X and Alzheimer's disease Y (but nothing else);

      iv. mortality is only affected by the unmeasured factors U in the diagram.

      In each of these cases, make clear if your conclusions differ depending on whether diabetes affects the risk of Alzheimer's disease or not.

Good luck!